

Monroe County (Rochester) Sample — Shopping Center / Retail



Traffic Impact Study

Prepared August 18, 2026 · 43.0640, -77.6150

SIS sample draft — full tier

Traffic Impact Study

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TRAFFIC IMPACT STUDY FOR DOT PROJECTS

Prepared in Accordance With Chapter 5 of the NYSDOT Highway Design Manual (HDM)

Project name	Monroe County (Rochester) Sample — Shopping Center / Retail
PIN	PIN xxxx.xx (to be assigned by NYSDOT)
Project description	Shopping Center / Retail (85 1,000 sqft GLA) within Rochester (NY) MSA.
NYSDOT Region	Region 4 — Rochester / Genesee Valley
Site coordinates	43.0640°, -77.6150°
Opening year (ETC)	2027

Note:

It is a violation of law for any person, unless they are acting under the direction of a licensed professional engineer, architect, landscape architect, or land surveyor, to alter an item in any way. If an item bearing the stamp of a licensed professional is altered, the altering engineer, architect, landscape architect, or land surveyor shall stamp the document and include the notation "altered by" followed by their signature, the date of such alteration, and a specific description of the alteration.

This report is based on the NYSDOT TIS Shell revised on 9/16/2014.

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PROJECT CONTEXT

NYSDOT region: Region 4 — Rochester / Genesee Valley**Planning group:** GTC (Genesee Transportation Council)**Regulatory regime:** SEQRA (ECL Article 8 / 6 NYCRR Part 617).**SEQRA classification:** Not yet determined — lead agency to classify as Type I (6 NYCRR Part 617.4 — likely EIS), Type II (Part 617.5 — exempt), or Unlisted (Part 617.6 — Short EAF + significance finding) before locking the TIS.

1.0 SUMMARY OF TRAFFIC IMPACTS

Capacity:

The proposed project will maintain an acceptable Level of Service of D or better in design year 2027.

Crash:

A minimum three-year accident history review was performed and the safety screening met all of the required steps. A full crash analysis is not required.

GML §239 referral: no state or county classified roadway was identified within 500 ft of the site via the NYSDOT RDM FeatureServer. The lead agency should still verify against the county's locally-adopted §239 mapping (some counties extend referral to additional facilities — e.g. county-designated scenic byways).

9	0	0	1.1s
INTERSECTIONS	LOS DROPS	AT LOS E/F	WORST Δ DELAY

2.0 TRAVEL SPEEDS

Per NYSDOT HDM Chapter 5 §5.2, posted speed limits and actual operating speeds (85th-percentile) along the study network are reported below. Posted-speed-limit values in the third column are auto-ingested from the NYSDOT Roadway Data Management (RDM) FeatureServer (https://gis.dot.ny.gov/hostingny/rest/services/Roadways/RDM_Roadway_Current/FeatureServer) at PDF-generation time — the same source data viewable through the NYSDOT Traffic Data Viewer (<https://gis.dot.ny.gov/>)

html5viewer/?viewer=tdv). Operating speeds (85th-percentile) are NOT published by NYSDOT and require a project-specific radar / floating-car speed study.

Street	Functional class	Posted speed	85th-pctile operating speed
Near Calkins Road (CALKINS RD)	Urban Minor Arterial	35 mph	Speed study required
East Henrietta Road	Urban Minor Arterial	35 mph	Speed study required
Near Calkins Road (CALKINS RD)	Urban Minor Arterial	35 mph	Speed study required
East Henrietta Road (CALKINS RD)	Urban Major Collector	35 mph	Speed study required
Near Calkins Road (CALKINS RD)	Urban Minor Arterial	35 mph	Speed study required
Calkins Road (CALKINS RD)	Urban Minor Arterial	35 mph	Speed study required

3.0 CAPACITY ANALYSIS

Capacity analyses performed in this report are consistent with HCM 6th Edition. The software used to perform this analysis is the SimpleImpactStudies HCM 6 solver (per-approach control-delay model). Synchro / SimTraffic, HCS, Sidra, or VISSIM may be substituted for formal submittal per NYSDOT Regional Traffic Office preference.

Study network functional-class composition (per NYSDOT RDM): Urban Minor Arterial (7); Urban Major Collector (1); Urban Principal Arterial Other (1). NYSDOT functional class drives access management expectations under HDM Appendix 5A (entrance standards for state highways) and the default K-factor banding under HDM §5.3.

Level of Service thresholds applied below are per HCM 2010 Exhibits 18-4 (signalized), 19-1 / 20-2 / 21-1 (unsignalized — 2-way stop / all-way stop / roundabout), and 10-7 (freeway). Any component $v/c > 1.0$ is LOS F regardless of delay or density.

Signalized intersections (HCM 2010 Exhibit 18-4, p. 18-6) — control delay sec/veh; any $v/c > 1.0$ is F.

LOS	A	B	C	D	E	F
sec/veh	≤ 10	10-20	20-35	35-55	55-80	> 80

Unsignalized — 2-way stop / all-way stop / roundabout (HCM 2010 Exhibits 19-1, 20-2, 21-1) — sec/veh; any $v/c > 1.0$ is F.

LOS	A	B	C	D	E	F
sec/veh	≤ 10	10-15	15-25	25-35	35-50	> 50

Freeway basic / weaving / merge-diverge (HCM 2010 Exhibit 10-7, p. 10-9) — density pc/mi/ln; any component $vd/c > 1.00$ is F.

LOS	A	B	C	D	E	F
pc/mi/ln	≤ 11	11-18	18-26	26-35	35-45	> 45

3.1 Growth Rates

Background traffic growth is applied at 1.5% per year over 1 year. This rate is the per-station median CAGR measured from the NYSDOT Traffic Monitoring AADT FeatureServer for Region 4 — Rochester / Genesee Valley: 0.01%/yr median across 1,892 count stations (rolling actual-count window approximately 2000-2019; 25th-75th percentile range -1.45% to 1.40%). The measured value supersedes the NYSDOT statewide default fallbacks (per HDM Chapter 5: 1.7%/yr Interstate + ramps from the Data Services Bureau, 2.0%/yr all other roads from the Regional Planning Group). The controlling Regional Planning Group GTC (Genesee Transportation Council) may have separately-adopted growth-rate guidance that overrides the measured median for formal submittal — confirm at scoping.

3.2 Existing AADT and DHV

Existing AADT is sourced from the NYSDOT Traffic Data Viewer. Design Hourly Volume (DHV) is reported per HDM Chapter 5 convention as $AADT \times K$ -factor; this analysis applies $K = 0.10$ as the

statewide design default — facility-specific K should be substituted from NYSDOT count-station records for formal submittal. The table below grows the existing volume at 1.5%/yr to Estimated Time of Completion (ETC), ETC + 20 years, and ETC + 30 years per HDM Chapter 5 §5.3.

Intersection	2026 AADT	2026 DHV	2027 (ETC) AADT	2027 (ETC) DHV	2047 AADT	2047 DHV	2057 AADT	2057 DHV
Near Calkins Road	10,000	1,000	10,150	1,015	13,671	1,367	15,865	1,587
East Henrietta Road & Lehigh Station Road	3,771	377	3,828	383	5,155	516	5,983	598
Near Calkins Road	10,000	1,000	10,150	1,015	13,671	1,367	15,865	1,587
East Henrietta Road & Calkins Road	9,720	972	9,866	987	13,288	1,329	15,421	1,542
Near Calkins Road	6,639	664	6,739	674	9,076	908	10,533	1,053
Calkins Road & Hylan Drive	6,639	664	6,739	674	9,076	908	10,533	1,053
Near East Henrietta Road	9,720	972	9,866	987	13,288	1,329	15,421	1,542
Near Hylan Drive	6,640	664	6,740	674	9,077	908	10,535	1,053
Middle Road & Calkins Road	7,830	783	7,947	795	10,704	1,070	12,423	1,242

Source: AADT estimated from engine peak-hour entering-volume sum via $K = 0.10$. NYSDOT Traffic Data Viewer measured AADT is recommended for submittal. Projection applies the 1.5%/yr background growth rate compounded annually.

3.3 Traffic Control Device Data

Existing traffic control devices at each study-network intersection are summarized below. Signal timing data (cycle length, phase splits, offset) and per-approach detection presence should be confirmed against the controlling NYSDOT Regional Traffic Office or municipal traffic-engineering signal record for formal submittal.

Intersection	Control type	Approaches w/ detection
Near Calkins Road	Signal (controlling RTO to confirm)	Field verification required
East Henrietta Road & Lehigh Station Road	Signal (controlling RTO to confirm)	Field verification required
Near Calkins Road	Signal (controlling RTO to confirm)	Field verification required
East Henrietta Road & Calkins Road	Signal (controlling RTO to confirm)	Field verification required
Near Calkins Road	Signal (controlling RTO to confirm)	Field verification required
Calkins Road & Hylan Drive	Signal (controlling RTO to confirm)	Field verification required
Near East Henrietta Road	Signal (controlling RTO to confirm)	Field verification required
Near Hylan Drive	Signal (controlling RTO to confirm)	Field verification required
Middle Road & Calkins Road	Signal (controlling RTO to confirm)	Field verification required

3.4 Capacity Analysis for Existing No-Build Condition

Existing No-Build conditions reflect current-year peak-hour volumes (no growth, no project trips). Existing measured volumes are grown to opening year 2027 at 1.5%/yr to establish the No-Build baseline for §3.5 comparison.

Intersection	Existing LOS	Existing delay (s)	No-Build LOS	No-Build delay (s)	v/c
Near Calkins Road	C	20.9	C	21.1	0.56
East Henrietta Road & Lehigh Station Road	B	15.6	B	15.7	0.21
Near Calkins Road	C	20.9	C	21.1	0.56
East Henrietta Road & Calkins Road	C	20.6	C	20.7	0.55
Near Calkins Road	B	17.6	B	17.7	0.37
Calkins Road & Hylan Drive	B	17.6	B	17.7	0.37

Intersection	Existing LOS	Existing delay (s)	No-Build LOS	No-Build delay (s)	v/c
Near East Henrietta Road	C	20.6	C	20.7	0.55
Near Hylan Drive	B	17.6	B	17.7	0.37
Middle Road & Calkins Road	B	18.6	B	18.7	0.44

Trip Distribution by Time of Day

Estimated within-day distribution of the 6,800 gross daily trips and the resulting on-site accumulation. Within-day distribution from NHTS 2022 shopping / errand / meal trip start times, cross-checked against public-data time-of-day patterns for LU 820 / 850. Activity builds to a midday plateau and a stronger late-afternoon/evening peak; on-site occupancy peaks midday through the evening.

Figure — Inbound / Outbound Trips by Start Time

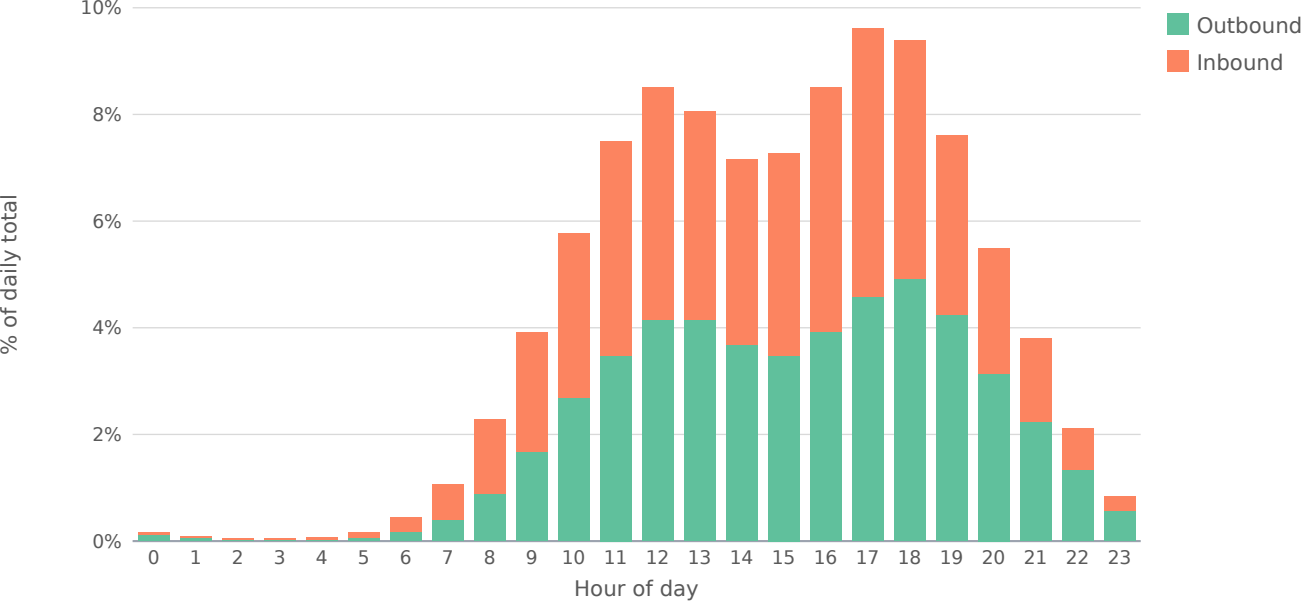
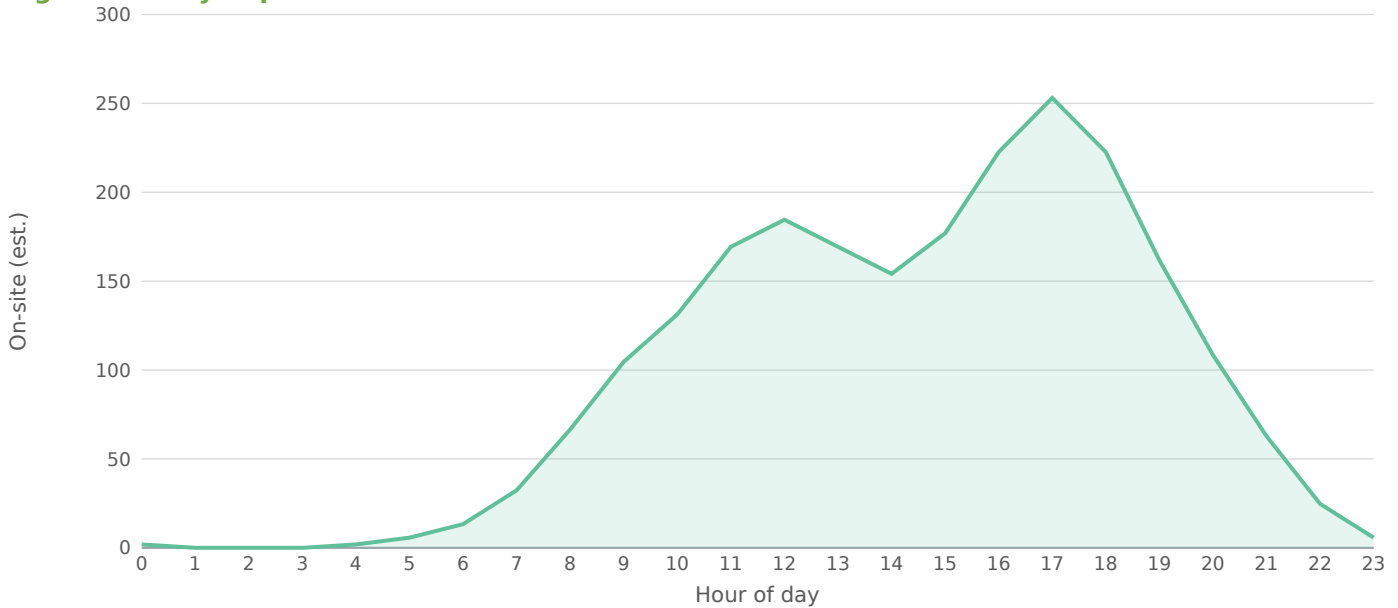


Figure — Daily Trip Accumulation



Peak on-site accumulation ~253 at 17:00, from 6,800 gross daily trips on the retail within-day profile. Screening estimate; not a substitute for a calibrated time-of-day model.

3.5 Capacity Analysis for Proposed Build Condition

Build conditions add the project's external peak-hour trips (680 PM peak, 326 inbound / 354 outbound) to the No-Build baseline at each affected intersection. Per HDM Chapter 5, intersection projects should also analyze each adjacent intersection plus any signal within ½ mile of the project site if expected left-turn volume changes materially affect capacity. The engine's network is determined by a 1.00-mile screening radius; manual scoping should confirm the ½-mile left-turn rule is satisfied.

Intersection	ETC No-Build	ETC Build	ETC+20 NB	ETC+20 Bld	Δ delay (s)
Near Calkins Road	C	C	C	C	1.1
East Henrietta Road & Lehigh Station Road	B	B	B	B	0.5
Near Calkins Road	C	C	C	C	0.7
East Henrietta Road & Calkins Road	C	C	C	C	0.5
Near Calkins Road	B	B	B	C	0.2
Calkins Road & Hylan Drive	B	B	B	B	0.1
Near East Henrietta Road	C	C	C	C	0.2
Near Hylan Drive	B	B	B	B	0.0
Middle Road & Calkins Road	B	B	C	C	0.1

3.6 Capacity Improvement Measures

No capacity improvements are required under Build conditions. The project will operate within acceptable LOS standards at all affected intersections per HDM Chapter 5. Standard HDM Chapter 5 improvement categories (turn lanes, signal installation / re-timing, signing and striping, roundabout conversion, TDM / TSM per HDM Chapter 24) remain available if site-specific concerns arise during NYSDOT Regional Traffic Office review.

3.7 Trip Distribution — Gravity Model

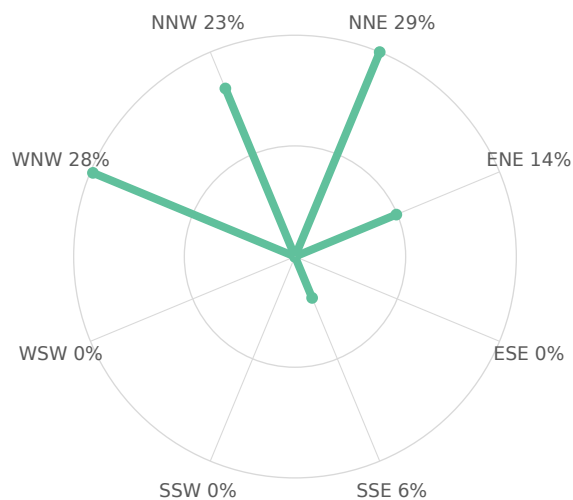
The trip distribution uses the gravity model (term = $M / (d^{1.1} \cdot d_{site})$; mass basis: intersection through-volume (AADT × K-factor) as the destination-activity attraction proxy). Shares are normalized to 100% of project trips and drive the trip assignment below. Basis: NCHRP-716 gamma-friction gravity model (production-constrained) over study-area attraction zones.

Directional sector	Share of project trips
Northeast (NNE + ENE)	43%
Southeast (ESE + SSE)	6%
Southwest (SSW + WSW)	0%
Northwest (WNW + NNW)	52%

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
Near Calkins Road	NNE	0.33	1,000	0.2	15.36%
Near Calkins Road	NNW	0.41	1,000	0.1	14.92%
East Henrietta Road & Calkins Road	ENE	0.49	972	0.1	14.13%
Near East Henrietta Road	NNE	0.71	972	0.1	13.14%
Middle Road & Calkins Road	WNW	0.93	783	0.1	9.85%
Near Calkins Road	WNW	0.60	664	0.1	9.32%
Calkins Road & Hylan Drive	WNW	0.70	664	0.1	9.00%
Near Hylan Drive	NNW	0.87	664	0.1	8.52%
East Henrietta Road & Lehigh Station Road	SSE	0.34	377	0.1	5.76%

9 study-area zone(s). Screening-grade distribution; not a calibrated regional model.

Figure — Directional Distribution of Project Trips



Screening-grade directional distribution of net new project trips by compass octant (spoke length $\propto$ percent of project trips). Derived from the Gravity Model distribution.

Figure — Project Trip Share by Study-Area Zone

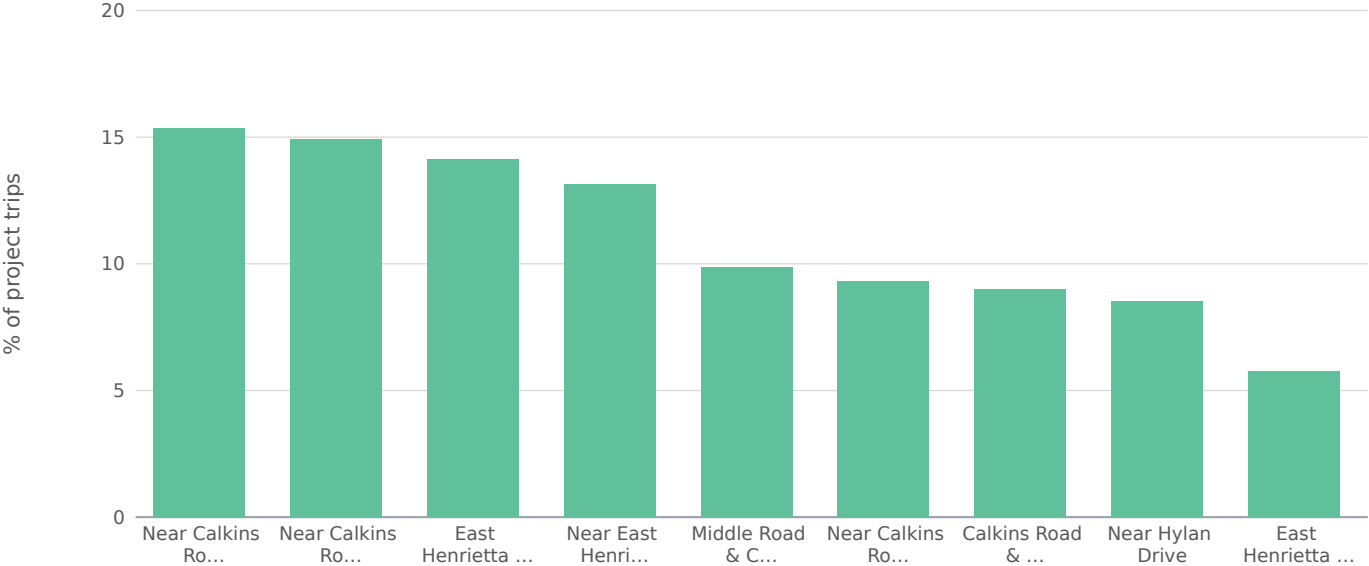
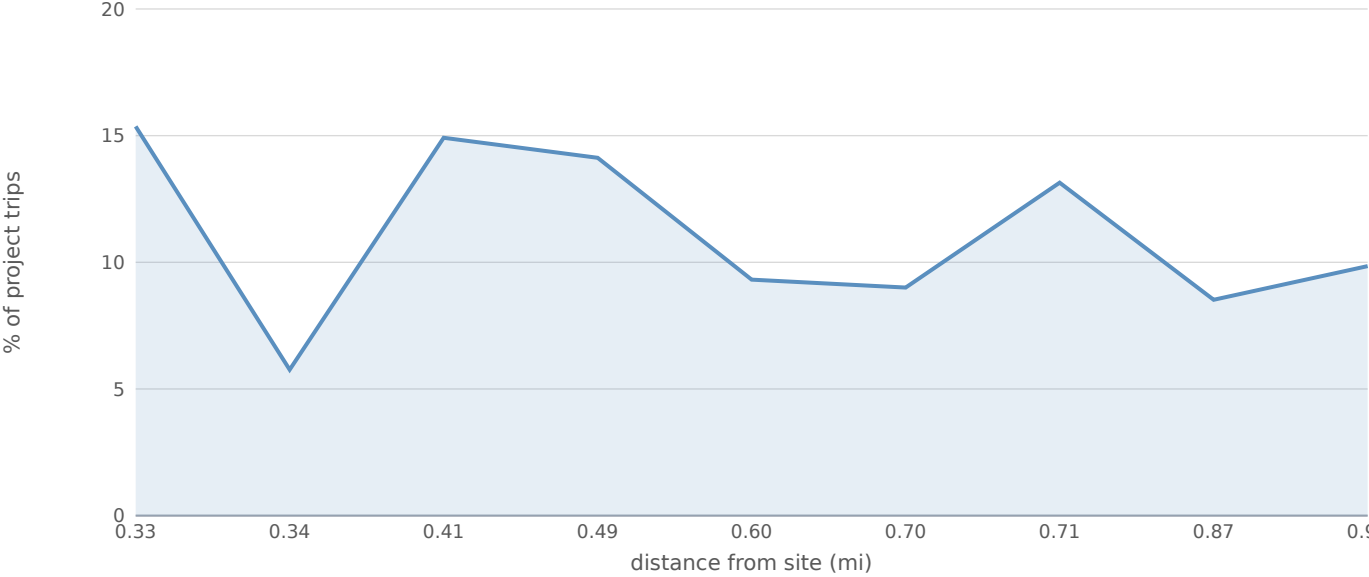


Figure — Trip Share vs. Distance from Site (Gravity Decay)



3.8 Project Trip Assignment

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
Near Calkins Road	NNE	0.33	46	86	26.8%
East Henrietta Road & Lehigh Station Road	SSE	0.34	42	80	24.9%
Near Calkins Road	NNW	0.41	30	57	17.8%
East Henrietta Road & Calkins Road	ENE	0.49	20	39	12.1%
Near Calkins Road	WNW	0.60	12	23	7.2%
Calkins Road & Hylan Drive	WNW	0.70	7	13	4.0%

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
Near East Henrietta Road	NNE	0.71	7	13	4.0%
Near Hylan Drive	NNW	0.87	3	6	1.9%
Middle Road & Calkins Road	WNW	0.93	2	4	1.2%

Project trips assigned to each study intersection from the distribution shares above: the directional shares orient the loading toward the high-share sectors, distance-decayed to each approach.

4.0 CRASH ANALYSIS

A minimum three-year accident history review was performed for the period 2023 to 2026 per HDM Chapter 5 §5.3.4. Auto-ingest from the New York State Police 3-year Crash Case Information dataset ([data.ny.gov / e8ky-4vqe](https://data.ny.gov/e8ky-4vqe)) was attempted but did not return county-level totals — either the project site's county could not be resolved via the NYSDOT RDM FeatureServer, or the SODA endpoint is unavailable. Refer to Region 4 — Rochester / Genesee Valley Regional Traffic Office for SIMS output documentation; the Safety Screening criteria from HDM Chapter 5 §5.3.4 (overall accident rate vs. 1.5× statewide average for comparable facility; HAL / PIL / SDL pattern flags; Fatal / Injury / Fatal+Injury rates not above average; addressed PIL list status; Fixed Object, Run-Off Road, and Wet-Road PIL list status) must be applied manually for this project.

APPENDIX A — EXISTING VOLUME REPORT

Per-approach existing peak-hour volumes (vph) for all affected intersections within study limits.
Source: SimpleImpactStudies HCM 6 solver, calibrated against the controlling DOT 511 / Regional Traffic Office data feed.

Intersection	Approach	Existing (vph)	No-Build (vph)	Build (vph)
Near Calkins Road	NB	288	293	338
Near Calkins Road	SB	297	301	334
Near Calkins Road	EB	238	242	245
Near Calkins Road	WB	177	180	185
East Henrietta Road & Lehigh Station Road	NB	110	112	150
East Henrietta Road & Lehigh Station Road	SB	93	94	136
East Henrietta Road & Lehigh Station Road	EB	99	100	100
East Henrietta Road & Lehigh Station Road	WB	75	76	76
Near Calkins Road	NB	360	366	395
Near Calkins Road	SB	227	230	246
Near Calkins Road	EB	227	230	242
Near Calkins Road	WB	186	189	189
East Henrietta Road & Calkins Road	NB	294	299	299
East Henrietta Road & Calkins Road	SB	233	236	249
East Henrietta Road & Calkins Road	EB	225	228	248
East Henrietta Road & Calkins Road	WB	220	223	228
Near Calkins Road	NB	185	188	188
Near Calkins Road	SB	177	180	185
Near Calkins Road	EB	181	183	189
Near Calkins Road	WB	121	123	135
Calkins Road & Hylan Drive	NB	209	213	213
Calkins Road & Hylan Drive	SB	144	147	149
Calkins Road & Hylan Drive	EB	171	173	177
Calkins Road & Hylan Drive	WB	139	142	149
Near East Henrietta Road	NB	319	324	331
Near East Henrietta Road	SB	234	238	243
Near East Henrietta Road	EB	222	226	226
Near East Henrietta Road	WB	196	199	201
Near Hylan Drive	NB	210	213	216
Near Hylan Drive	SB	151	153	155
Near Hylan Drive	EB	152	154	155
Near Hylan Drive	WB	152	154	154
Middle Road & Calkins Road	NB	228	231	231
Middle Road & Calkins Road	SB	207	210	211
Middle Road & Calkins Road	EB	170	173	174

Intersection	Approach	Existing (vph)	No-Build (vph)	Build (vph)
Middle Road & Calkins Road	WB	178	181	183

APPENDIX B — EXISTING CONDITION CAPACITY ANALYSIS OUTPUT

Per-intersection HCM 6 solver output for the Existing No-Build condition (current-year volumes; volumes grown to opening year at the §3.1 background-growth rate for No-Build comparison).

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Near Calkins Road	NB	293	0.36	17.5	B
Near Calkins Road	SB	301	0.37	17.7	B
Near Calkins Road	EB	242	0.30	16.7	B
Near Calkins Road	WB	180	0.22	15.8	B
East Henrietta Road & Lehigh Station Road	NB	112	0.14	14.9	B
East Henrietta Road & Lehigh Station Road	SB	94	0.12	14.7	B
East Henrietta Road & Lehigh Station Road	EB	100	0.12	14.7	B
East Henrietta Road & Lehigh Station Road	WB	76	0.09	14.4	B
Near Calkins Road	NB	366	0.45	18.9	B
Near Calkins Road	SB	230	0.28	16.5	B
Near Calkins Road	EB	230	0.28	16.5	B
Near Calkins Road	WB	189	0.23	15.9	B
East Henrietta Road & Calkins Road	NB	299	0.37	17.6	B
East Henrietta Road & Calkins Road	SB	236	0.29	16.6	B
East Henrietta Road & Calkins Road	EB	228	0.28	16.5	B
East Henrietta Road & Calkins Road	WB	223	0.28	16.4	B
Near Calkins Road	NB	188	0.23	15.9	B
Near Calkins Road	SB	180	0.22	15.8	B
Near Calkins Road	EB	183	0.23	15.8	B
Near Calkins Road	WB	123	0.15	15.0	B
Calkins Road & Hylan Drive	NB	213	0.26	16.2	B
Calkins Road & Hylan Drive	SB	147	0.18	15.3	B
Calkins Road & Hylan Drive	EB	173	0.21	15.7	B
Calkins Road & Hylan Drive	WB	142	0.17	15.2	B
Near East Henrietta Road	NB	324	0.40	18.1	B
Near East Henrietta Road	SB	238	0.29	16.6	B
Near East Henrietta Road	EB	226	0.28	16.4	B
Near East Henrietta Road	WB	199	0.25	16.0	B
Near Hylan Drive	NB	213	0.26	16.2	B
Near Hylan Drive	SB	153	0.19	15.4	B
Near Hylan Drive	EB	154	0.19	15.4	B
Near Hylan Drive	WB	154	0.19	15.4	B
Middle Road & Calkins Road	NB	231	0.29	16.5	B
Middle Road & Calkins Road	SB	210	0.26	16.2	B
Middle Road & Calkins Road	EB	173	0.21	15.7	B
Middle Road & Calkins Road	WB	181	0.22	15.8	B

APPENDIX C — PROPOSED CONDITION CAPACITY ANALYSIS OUTPUT

Per-intersection HCM 6 solver output for the Proposed Build condition (No-Build volumes plus project external trips at the assigned distribution).

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Near Calkins Road	NB	338	0.42	18.3	B
Near Calkins Road	SB	334	0.41	18.3	B
Near Calkins Road	EB	245	0.30	16.7	B
Near Calkins Road	WB	185	0.23	15.8	B
East Henrietta Road & Lehigh Station Road	NB	150	0.19	15.4	B
East Henrietta Road & Lehigh Station Road	SB	136	0.17	15.2	B
East Henrietta Road & Lehigh Station Road	EB	100	0.12	14.7	B
East Henrietta Road & Lehigh Station Road	WB	76	0.09	14.4	B
Near Calkins Road	NB	395	0.49	19.5	B
Near Calkins Road	SB	246	0.30	16.7	B
Near Calkins Road	EB	242	0.30	16.7	B
Near Calkins Road	WB	189	0.23	15.9	B
East Henrietta Road & Calkins Road	NB	299	0.37	17.6	B
East Henrietta Road & Calkins Road	SB	249	0.31	16.8	B
East Henrietta Road & Calkins Road	EB	248	0.31	16.8	B
East Henrietta Road & Calkins Road	WB	228	0.28	16.5	B
Near Calkins Road	NB	188	0.23	15.9	B
Near Calkins Road	SB	185	0.23	15.8	B
Near Calkins Road	EB	189	0.23	15.9	B
Near Calkins Road	WB	135	0.17	15.2	B
Calkins Road & Hylan Drive	NB	213	0.26	16.2	B
Calkins Road & Hylan Drive	SB	149	0.18	15.3	B
Calkins Road & Hylan Drive	EB	177	0.22	15.7	B
Calkins Road & Hylan Drive	WB	149	0.18	15.3	B
Near East Henrietta Road	NB	331	0.41	18.2	B
Near East Henrietta Road	SB	243	0.30	16.7	B
Near East Henrietta Road	EB	226	0.28	16.4	B
Near East Henrietta Road	WB	201	0.25	16.0	B
Near Hylan Drive	NB	216	0.27	16.3	B
Near Hylan Drive	SB	155	0.19	15.4	B
Near Hylan Drive	EB	155	0.19	15.4	B
Near Hylan Drive	WB	154	0.19	15.4	B
Middle Road & Calkins Road	NB	231	0.29	16.5	B
Middle Road & Calkins Road	SB	211	0.26	16.2	B
Middle Road & Calkins Road	EB	174	0.21	15.7	B

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Middle Road & Calkins Road	WB	183	0.23	15.8	B

APPENDIX D — CRASH ANALYSIS DIAGRAMS AND TABLES

Crash analysis diagrams and tables are not produced by this screening analysis. When the §4.0 crash-analysis section is expanded for formal submittal, the following NYSDOT forms are required: TE-156a (Collision Diagram), TE-164a (Safety Benefits Evaluation Form), TE-204a (Accident Rate Summary), and TE-213 (HAL / PIL / SDL Screening Output). SIMS output and statewide-average rate references should be obtained from the controlling Regional Traffic Office per HDM Chapter 5 §5.3.4 and the NYSDOT Office of Modal Safety statewide rate tables (<https://www.dot.ny.gov/divisions/operating/osss/highway/accident-rates>).

This report is based on the NYSDOT TIS Shell revised on 9/16/2014.

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	85 1,000 sqft GLA
Daily trips (productions)	6,800
PM peak-hour trips	680 (326 in / 354 out)

Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals: $T_j = P \cdot (A_j \cdot F_j) / \sum (A_x \cdot F_x)$. Attractiveness A_j is each signal's through-volume; the friction factor F_j is the NCHRP-716 gamma function $F = a \cdot t^b \cdot e^{(c \cdot t)}$ (home-based-work: $a=28507$, $b=-0.02$, $c=-0.123$) on the travel time t to the signal.

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
Near Calkins Road	0.33	0.8	86	26.8%
East Henrietta Road & Lehigh Station Road	0.34	0.8	80	24.9%
Near Calkins Road	0.41	1.0	57	17.8%
East Henrietta Road & Calkins Road	0.49	1.2	39	12.1%
Near Calkins Road	0.60	1.4	23	7.2%
Calkins Road & Hylan Drive	0.70	1.7	13	4.0%
Near East Henrietta Road	0.71	1.7	13	4.0%
Near Hylan Drive	0.87	2.1	6	1.9%
Middle Road & Calkins Road	0.93	2.2	4	1.2%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 94% applied via a binary logit $P(\text{auto}) = 1 / (1 + e^{-(ASC - \lambda \cdot \Delta GC)})$ calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 6% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

A capacity-constrained assignment using the BPR volume-delay function $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$ iteratively shifts trips away from over-capacity signals toward less-congested alternatives until the assignment is congestion-consistent. Highest post-build v/c among study signals: 0.61. Per-signal v/c , delay, LOS, and queue are in the capacity appendix. (Road-network corridor routing was not available for this region; per-signal capacity-constrained assignment was used.)

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

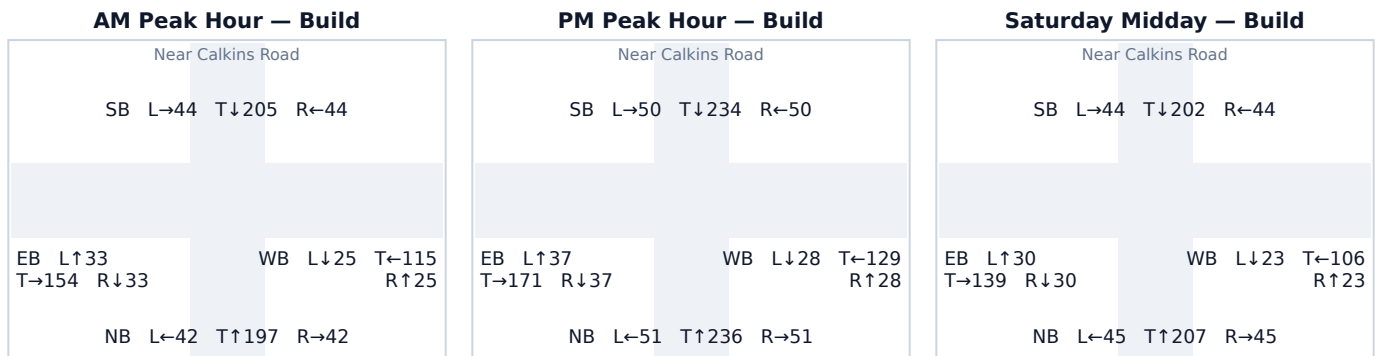
One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach HCM capacity table. Analysis follows the Highway Capacity Manual 6th Edition signalized-intersection method; v/c, control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

Background turning-movement volumes in the diagrams are distributed from each approach total using an estimated 15/70/15 (Left/Through/Right) split. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

A.1 Near Calkins Road

Signal ID	rochester-ny-418
Location	SE Rochester (NY) · 0.33 mi from site
Added PM peak trips	86
Existing / No-Build (PM)	LOS C · 21.1 s/veh · v/c 0.56
Build (PM)	LOS C · 22.2 s/veh · v/c 0.61
Δ control delay	+1.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	293	45	338	0.36 → 0.42	17.5 → 18.3	B → B	236
SB	301	32	334	0.37 → 0.41	17.7 → 18.3	B → B	232
EB	242	3	245	0.30 → 0.30	16.7 → 16.7	B → B	161
WB	180	6	185	0.22 → 0.23	15.8 → 15.8	B → B	117

Affected movements (PM peak project trips)

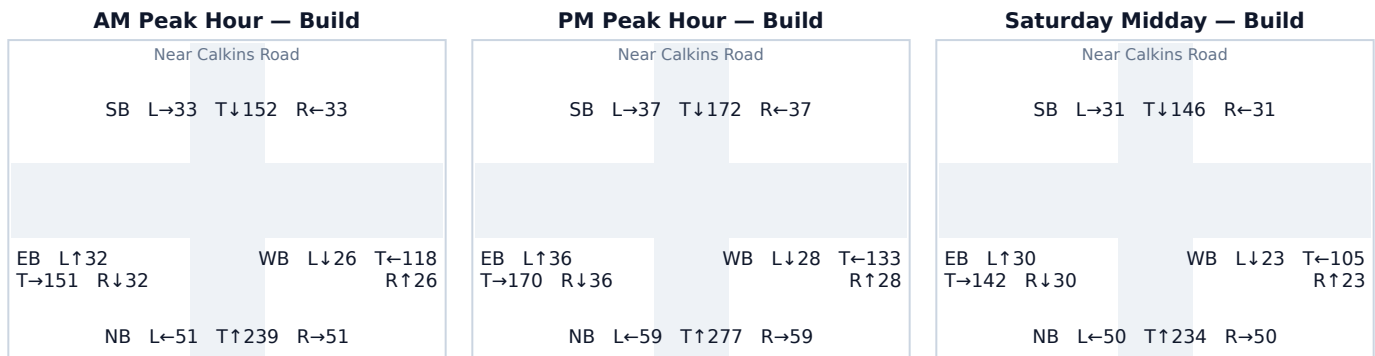
Movement	Project trips	% of project trips
NB Through	35	41%
SB Through	32	37%
NB Right	6	7%
WB Left	6	7%
NB Left	4	5%
EB Right	3	3%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.2 Near Calkins Road

Signal ID	rochester-ny-752
Location	SE Rochester (NY) · 0.41 mi from site
Added PM peak trips	57
Existing / No-Build (PM)	LOS C · 21.1 s/veh · v/c 0.56
Build (PM)	LOS C · 21.8 s/veh · v/c 0.60
Δ control delay	+0.7 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	366	29	395	0.45 → 0.49	18.9 → 19.5	B → B	287
SB	230	16	246	0.28 → 0.30	16.5 → 16.7	B → B	162
EB	230	12	242	0.28 → 0.30	16.5 → 16.7	B → B	158
WB	189	0	189	0.23 → 0.23	15.9 → 15.9	B → B	119

Affected movements (PM peak project trips)

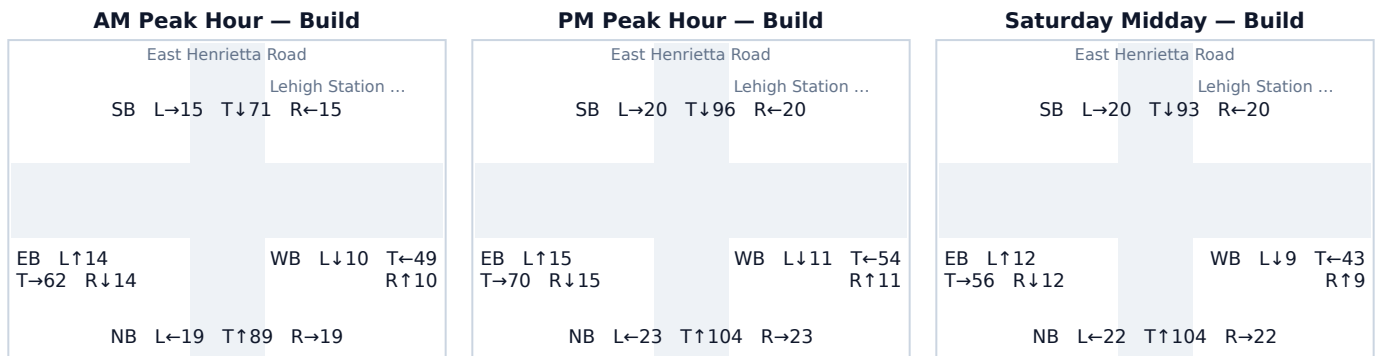
Movement	Project trips	% of project trips
NB Through	17	30%
SB Through	16	28%
NB Left	12	21%
EB Right	12	21%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.3 East Henrietta Road & Lehigh Station Road

Signal ID	rochester-ny-763
Location	SE Rochester (NY) · 0.34 mi from site
Added PM peak trips	80
Existing / No-Build (PM)	LOS B · 15.7 s/veh · v/c 0.21
Build (PM)	LOS B · 16.2 s/veh · v/c 0.26
Δ control delay	+0.5 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	112	38	150	0.14 → 0.19	14.9 → 15.4	B → B	93
SB	94	42	136	0.12 → 0.17	14.7 → 15.2	B → B	83
EB	100	0	100	0.12 → 0.12	14.7 → 14.7	B → B	60
WB	76	0	76	0.09 → 0.09	14.4 → 14.4	B → B	45

Affected movements (PM peak project trips)

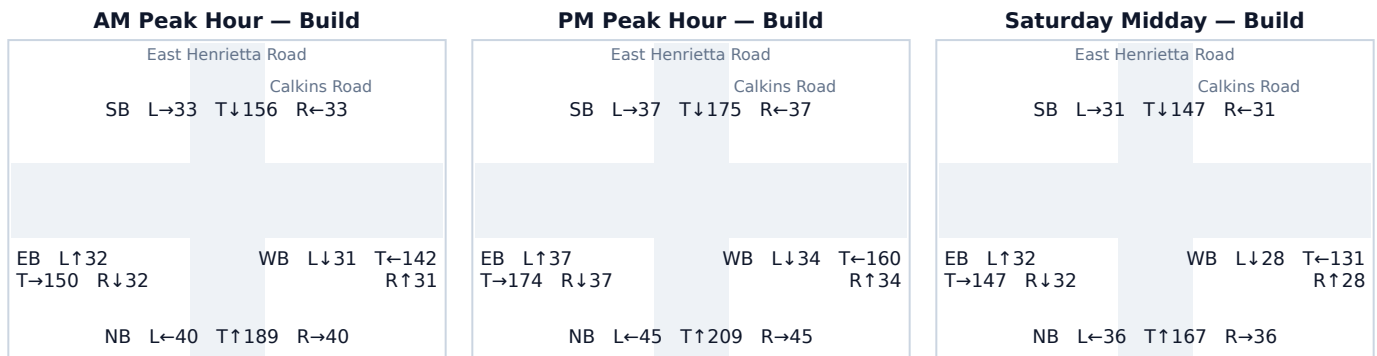
Movement	Project trips	% of project trips
SB Through	42	53%
NB Through	38	48%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.4 East Henrietta Road & Calkins Road

Signal ID	rochester-ny-753
Location	SE Rochester (NY) · 0.49 mi from site
Added PM peak trips	39
Existing / No-Build (PM)	LOS C · 20.7 s/veh · v/c 0.55
Build (PM)	LOS C · 21.2 s/veh · v/c 0.57
Δ control delay	+0.5 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	299	0	299	0.37 → 0.37	17.6 → 17.6	B → B	203
SB	236	13	249	0.29 → 0.31	16.6 → 16.8	B → B	164
EB	228	21	248	0.28 → 0.31	16.5 → 16.8	B → B	164
WB	223	5	228	0.28 → 0.28	16.4 → 16.5	B → B	148

Affected movements (PM peak project trips)

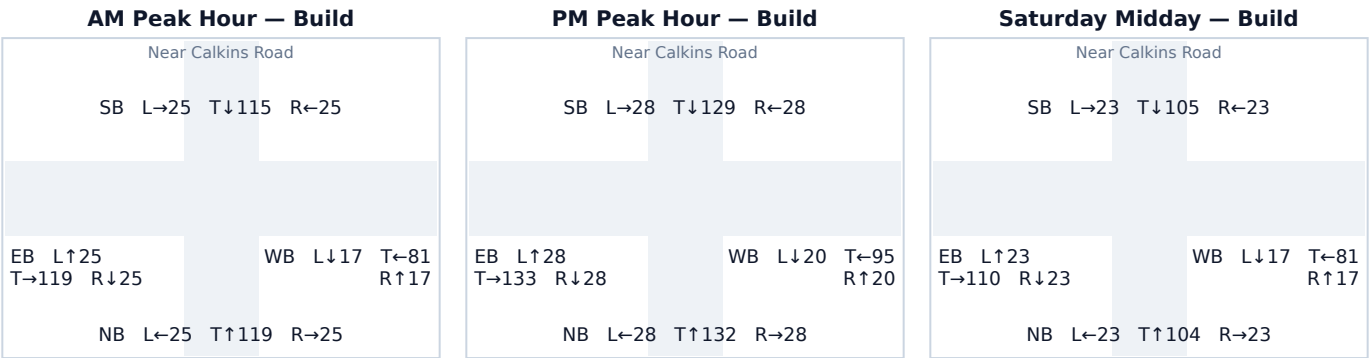
Movement	Project trips	% of project trips
EB Left	15	38%
SB Right	13	33%
EB Through	6	15%
WB Through	5	13%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.5 Near Calkins Road

Signal ID	rochester-ny-711
Location	SE Rochester (NY) · 0.60 mi from site
Added PM peak trips	23
Existing / No-Build (PM)	LOS B · 17.7 s/veh · v/c 0.37
Build (PM)	LOS B · 17.9 s/veh · v/c 0.39
Δ control delay	+0.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	188	0	188	0.23 → 0.23	15.9 → 15.9	B → B	119
SB	180	5	185	0.22 → 0.23	15.8 → 15.8	B → B	117
EB	183	6	189	0.23 → 0.23	15.8 → 15.9	B → B	120
WB	123	12	135	0.15 → 0.17	15.0 → 15.2	B → B	83

Affected movements (PM peak project trips)

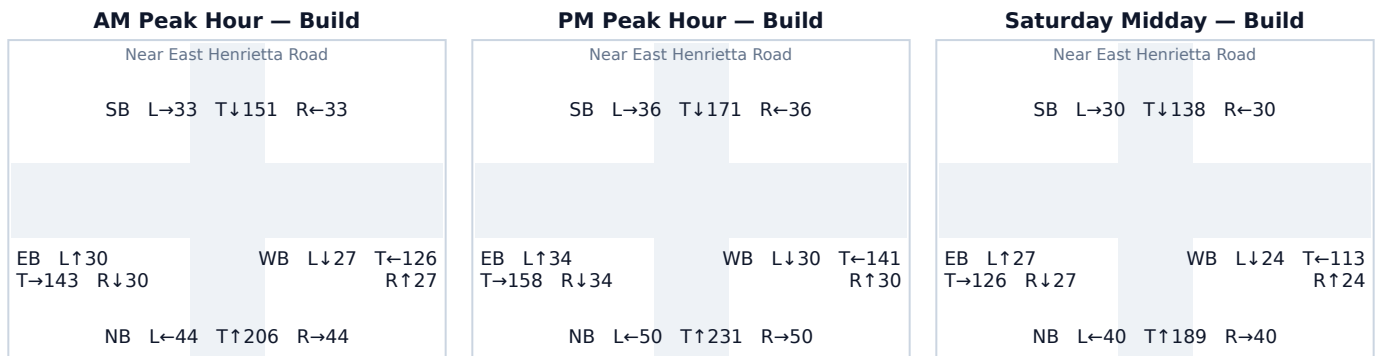
Movement	Project trips	% of project trips
WB Through	7	30%
EB Through	6	26%
WB Right	5	22%
SB Left	5	22%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.6 Near East Henrietta Road

Signal ID	rochester-ny-765
Location	SE Rochester (NY) · 0.71 mi from site
Added PM peak trips	13
Existing / No-Build (PM)	LOS C · 20.7 s/veh · v/c 0.55
Build (PM)	LOS C · 20.9 s/veh · v/c 0.56
Δ control delay	+0.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	324	7	331	0.40 → 0.41	18.1 → 18.2	B → B	230
SB	238	5	243	0.29 → 0.30	16.6 → 16.7	B → B	159
EB	226	0	226	0.28 → 0.28	16.4 → 16.4	B → B	146
WB	199	1	201	0.25 → 0.25	16.0 → 16.0	B → B	128

Affected movements (PM peak project trips)

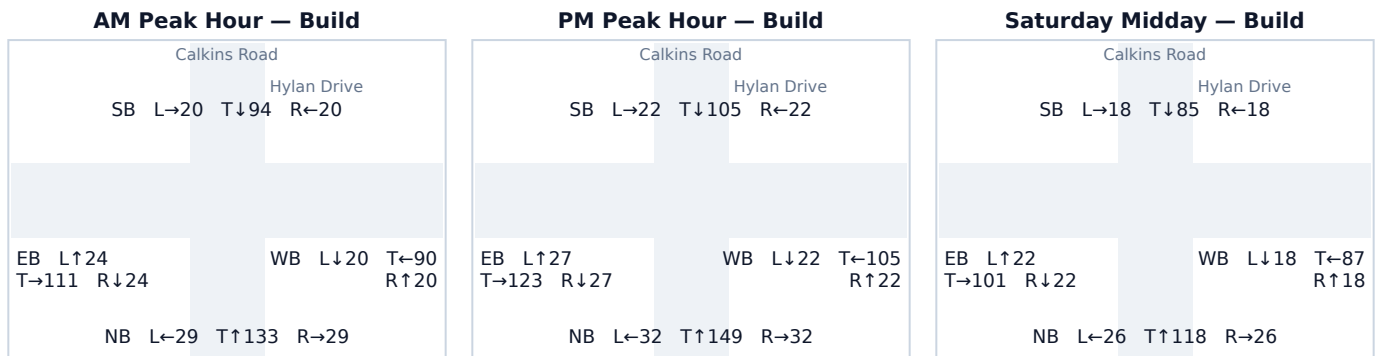
Movement	Project trips	% of project trips
NB Through	5	38%
SB Through	5	38%
NB Right	2	15%
WB Left	1	8%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.7 Calkins Road & Hylan Drive

Signal ID	rochester-ny-751
Location	SE Rochester (NY) · 0.70 mi from site
Added PM peak trips	13
Existing / No-Build (PM)	LOS B · 17.7 s/veh · v/c 0.37
Build (PM)	LOS B · 17.8 s/veh · v/c 0.38
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	213	0	213	0.26 → 0.26	16.2 → 16.2	B → B	137
SB	147	2	149	0.18 → 0.18	15.3 → 15.3	B → B	92
EB	173	4	177	0.21 → 0.22	15.7 → 15.7	B → B	111
WB	142	7	149	0.17 → 0.18	15.2 → 15.3	B → B	92

Affected movements (PM peak project trips)

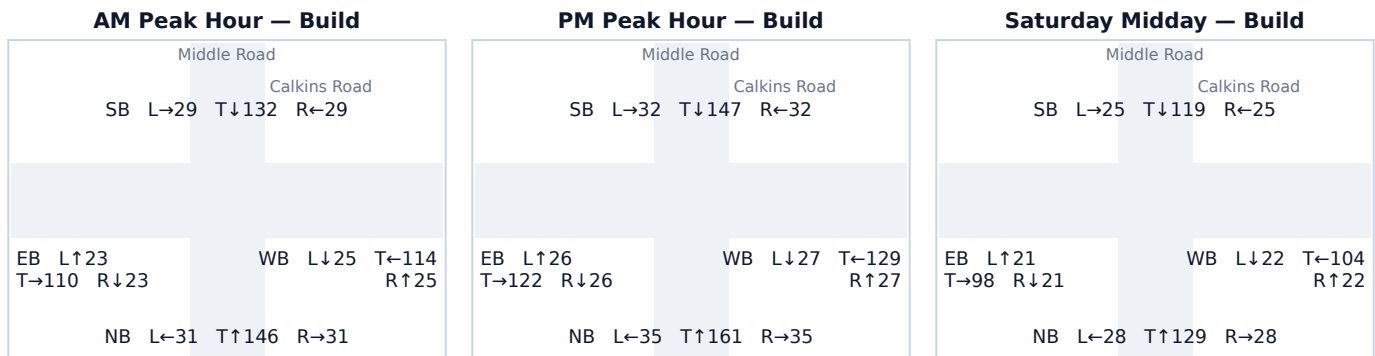
Movement	Project trips	% of project trips
WB Through	4	31%
EB Through	4	31%
WB Right	3	23%
SB Left	2	15%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.8 Middle Road & Calkins Road

Signal ID	rochester-ny-719
Location	SE Rochester (NY) · 0.93 mi from site
Added PM peak trips	4
Existing / No-Build (PM)	LOS B · 18.7 s/veh · v/c 0.44
Build (PM)	LOS B · 18.8 s/veh · v/c 0.44
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	231	0	231	0.29 → 0.29	16.5 → 16.5	B → B	151
SB	210	1	211	0.26 → 0.26	16.2 → 16.2	B → B	136
EB	173	1	174	0.21 → 0.21	15.7 → 15.7	B → B	109
WB	181	2	183	0.22 → 0.23	15.8 → 15.8	B → B	115

Affected movements (PM peak project trips)

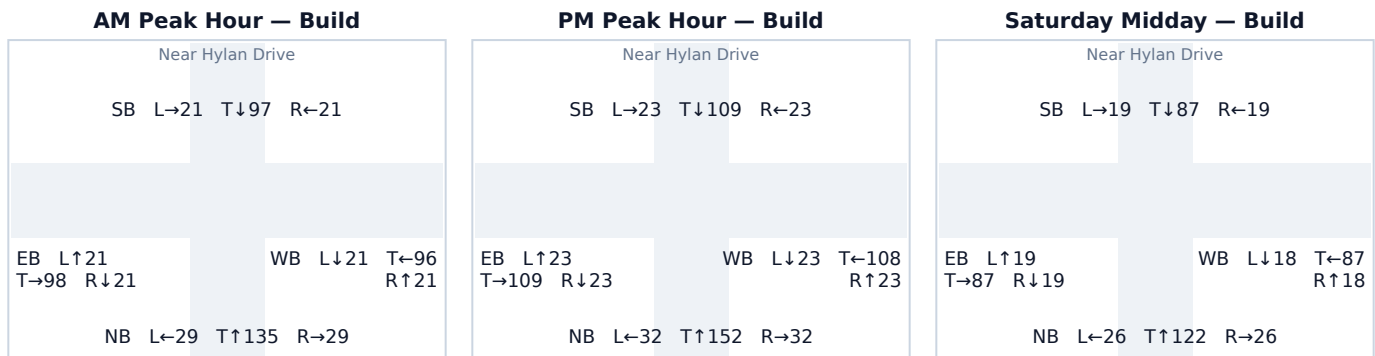
Movement	Project trips	% of project trips
WB Through	1	25%
EB Through	1	25%
WB Right	1	25%
SB Left	1	25%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.9 Near Hylan Drive

Signal ID	rochester-ny-906
Location	SE Rochester (NY) · 0.87 mi from site
Added PM peak trips	6
Existing / No-Build (PM)	LOS B · 17.7 s/veh · v/c 0.37
Build (PM)	LOS B · 17.7 s/veh · v/c 0.38
Δ control delay	+0.0 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	213	3	216	0.26 → 0.27	16.2 → 16.3	B → B	139
SB	153	2	155	0.19 → 0.19	15.4 → 15.4	B → B	96
EB	154	1	155	0.19 → 0.19	15.4 → 15.4	B → B	96
WB	154	0	154	0.19 → 0.19	15.4 → 15.4	B → B	95

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	2	33%
SB Through	2	33%
NB Left	1	17%
EB Right	1	17%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.